Firewall Sandwich

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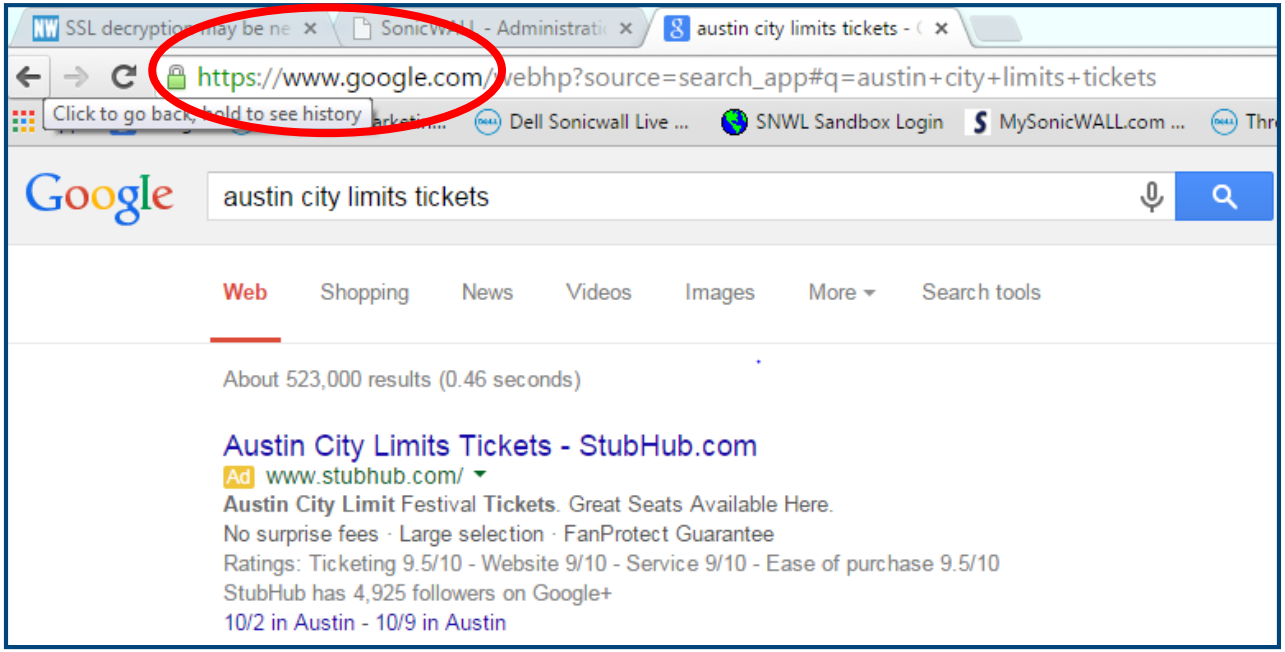
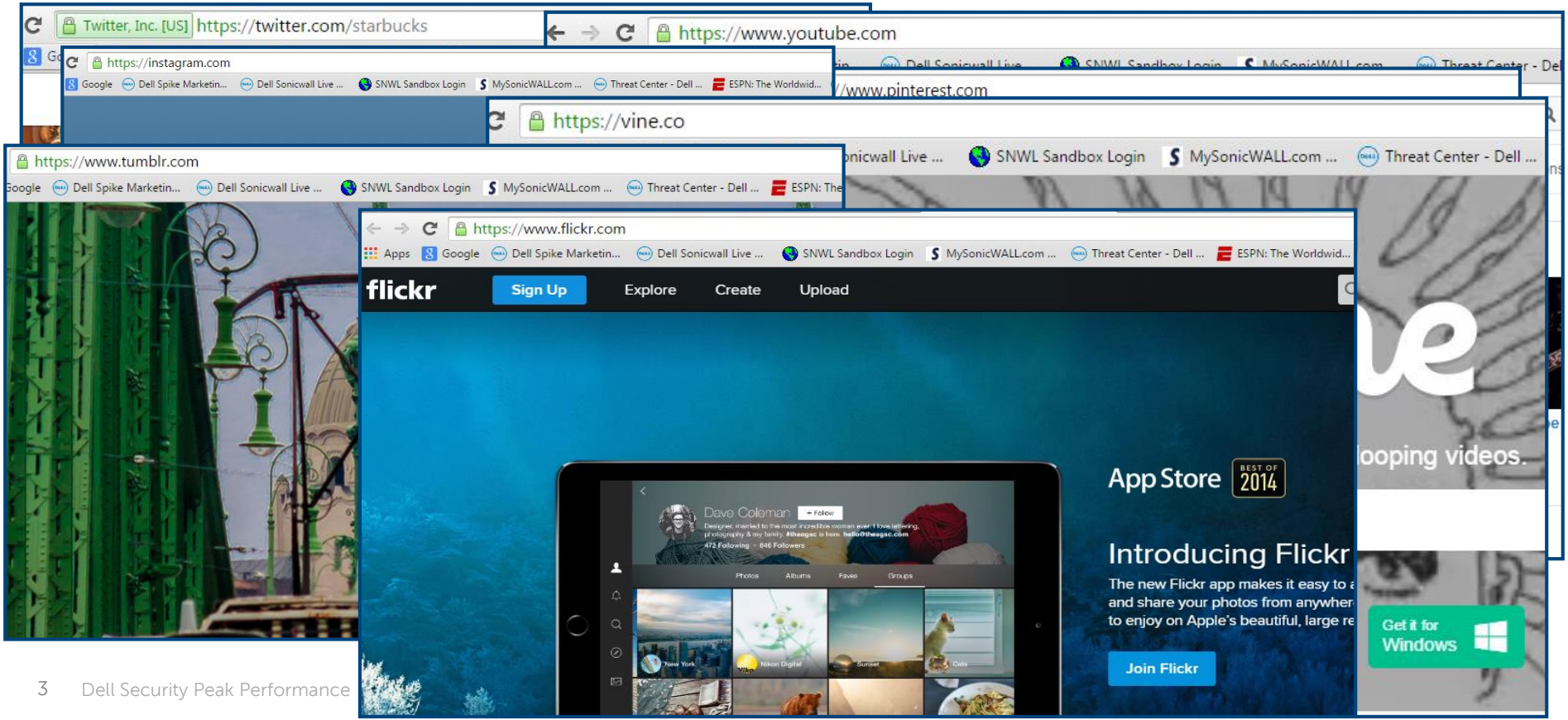
Many of your users' web sessions are **encrypted with HTTPS**

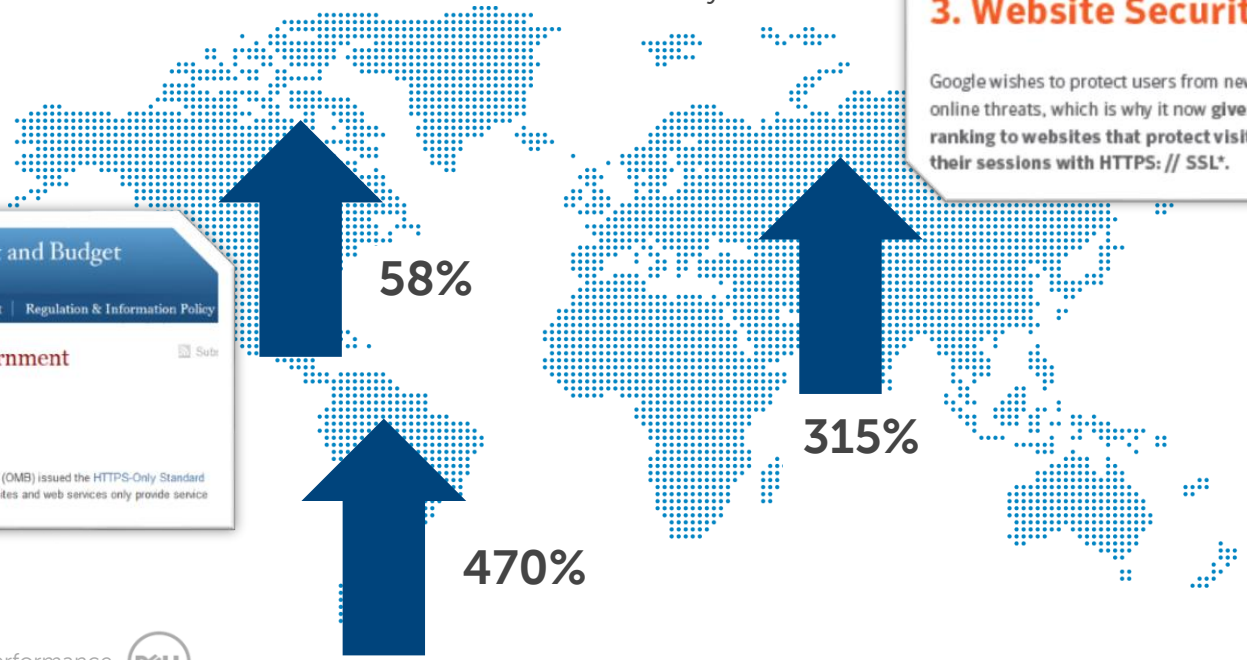


Many of your users' web sessions are **encrypted with HTTPS** – and so is the malware that targets them



Encrypted web traffic growth

- SSL/TLS comprises 15–20 percent of total web traffic and 25–35 percent of typical enterprise traffic
- There's an average yearly increase of 20 percent in SSL/TLS traffic
- Only 20 percent of enterprises with next-generation firewalls (NGFWs) inspect inbound/outbound SSL/TLS traffic
- 50 percent of all inbound/outbound attacks will use SSL/TLS by 2017

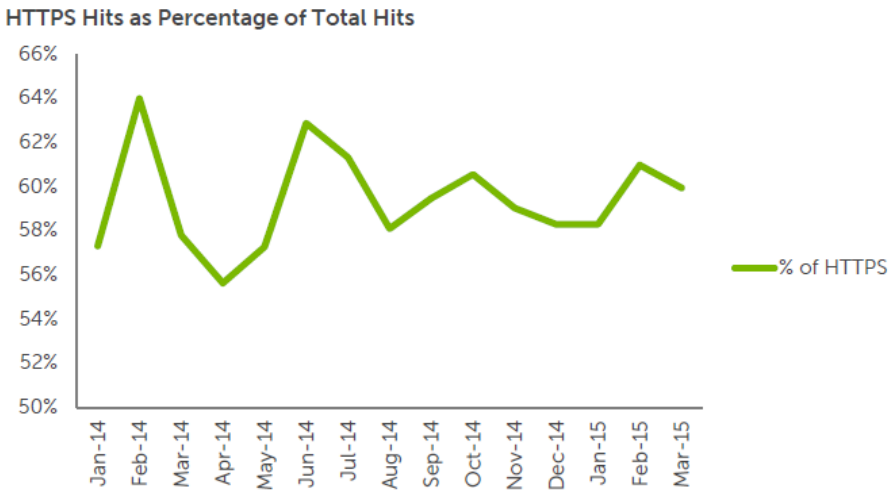


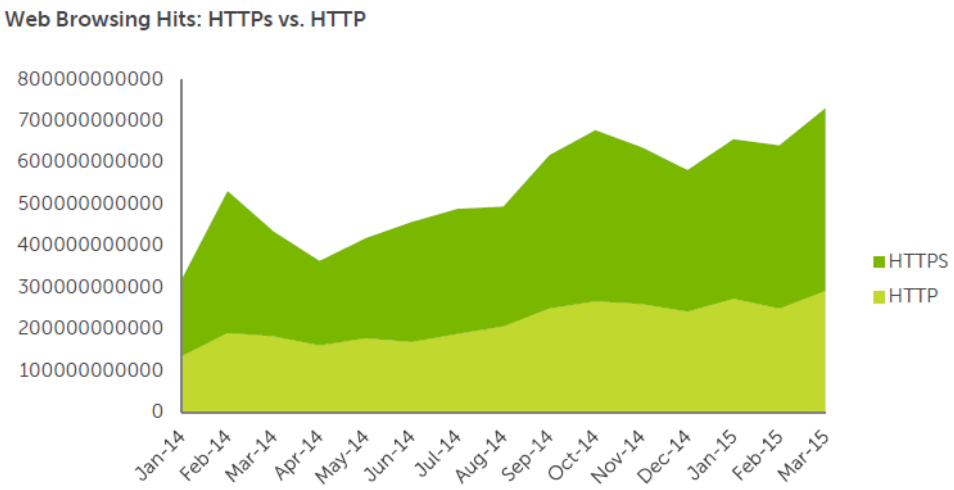
3. Website Security

Google wishes to protect users from new and evolving online threats, which is why it now **gives a higher SEO ranking to websites that protect visitors throughout their sessions with HTTPS: // SSL*.**

You can't protect what you can't see — hackers are hiding in plain sight (SSL/TLS encrypted traffic)

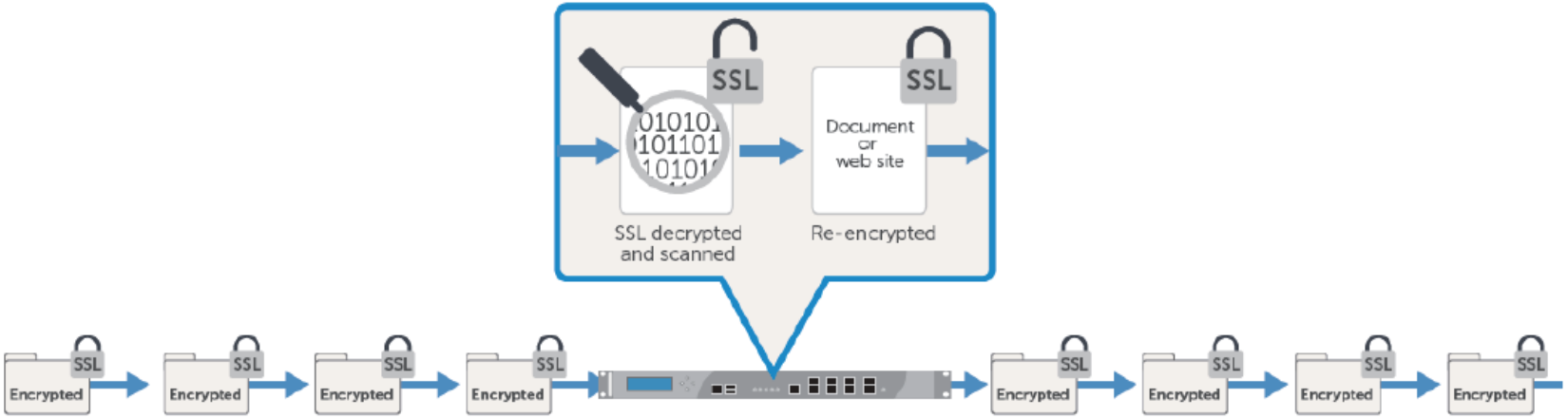
Dell saw HTTPS web connections grow 109 percent in CY2014





Example: Yahoo (SSL encrypted HTTPS sessions) third-party banner ads distributed malware over four days (27,000 users per hour) — an attack unseen by most firewalls

Organizations need next-generation protection to eliminate blind spots in SSL traffic — how much of your network traffic is HTTPS?



Next-generation firewalls intercept the keys exchanged during the SSL process, and then open and decrypt SSL traffic to look for threats. Network traffic is inspected for nefarious activity, before being re-encrypted and sent to the recipient.

How can we add and scale this critical protection to prevent lag and network latency?

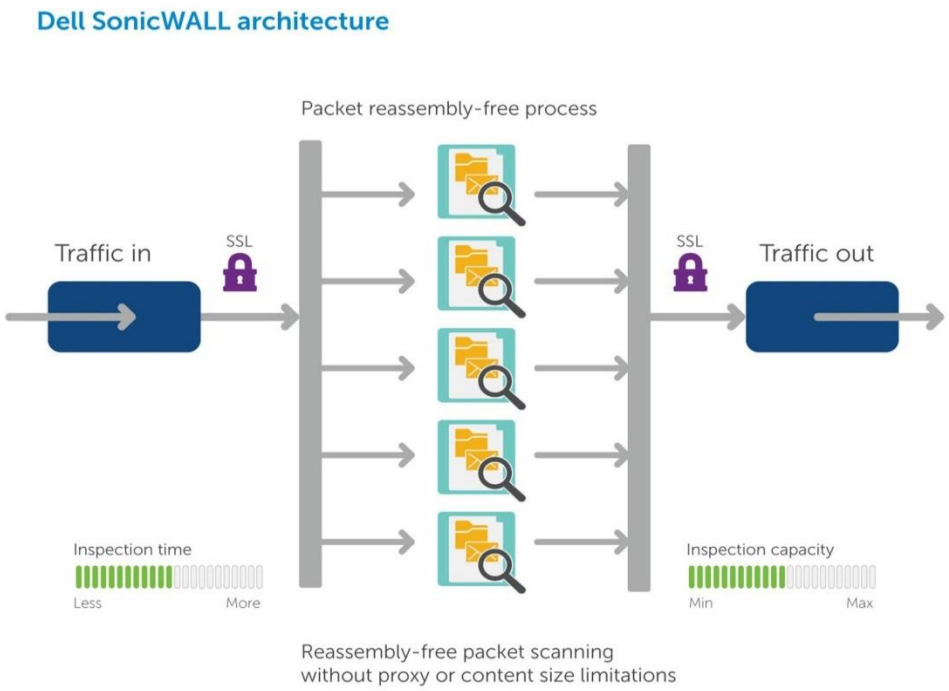
What are the industry limitations today?

- Processing power: key sizes, ciphers
- Knowledge of PKI, deployment pain
- Non-browser-based applications that leverage SSL (mobile, certain desktop apps)
- Distribution of certs in non-managed/trusted environments
- Connection count (memory allocation)
- Bypassing sites (whitelisting strategies)

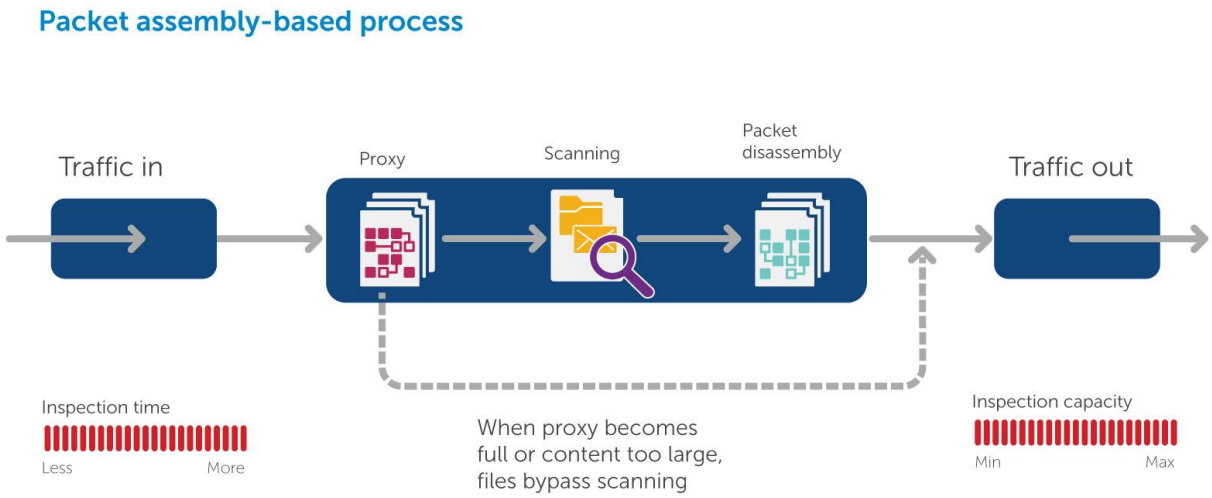
Introducing a super solution for massive security issues

Dell SonicWALL SuperMassive 9000 series powers next-generation enterprise solutions that provide SSL decryption, inspection and protection with no added latency

Reassembly-Free Deep Packet Inspection (RFDPI) vs. packet assembly-based architecture

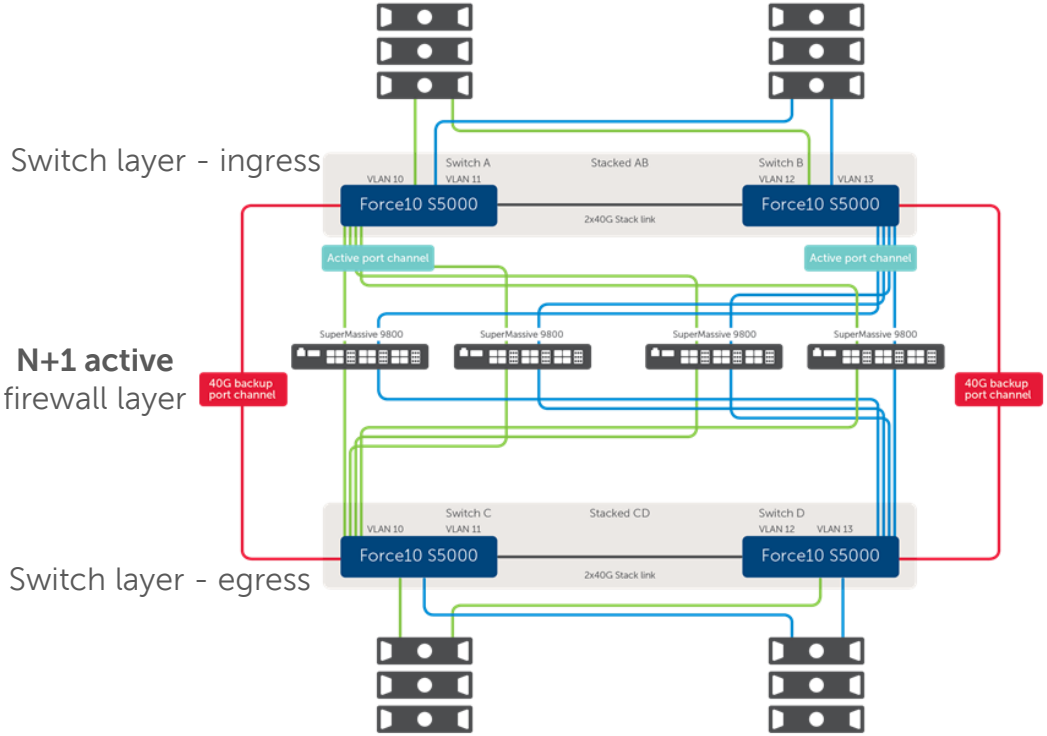


RFDPI vs. packet assembly-based architecture



Firewall sandwich architecture

- Provides the blueprint for deploying a network-based, scale-out security layer architecture
- Offers transparent security services via highly resilient Layer 2 design to enhance existing security solutions, separate security functions or provide added capacity
- Fully validates with Dell Networking S4810, S5000, S6000 and Dell Network Security SuperMassive 9x00 series NGFW products
- Provides N+1 redundancy (vs. 1+1) without reliance on HA or clustering protocols
- Supports 1, 10 or 40GE ingress/egress connections (today) and performance
- Utilizes VLT, symmetric hashing, port channel LAG, active mesh



Why such a design?

- Networking reasons
  - Scalability, commoditization of 10/40 GbE
  - Convergence of layers
  - Redundancy and resiliency
  - **Increases in east to west traffic**
    - **Virtualization, Datacenters**

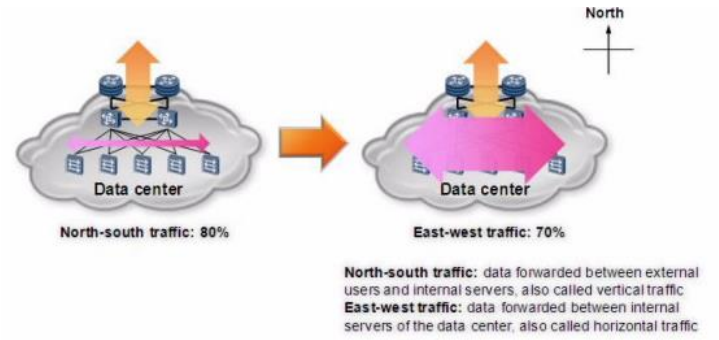
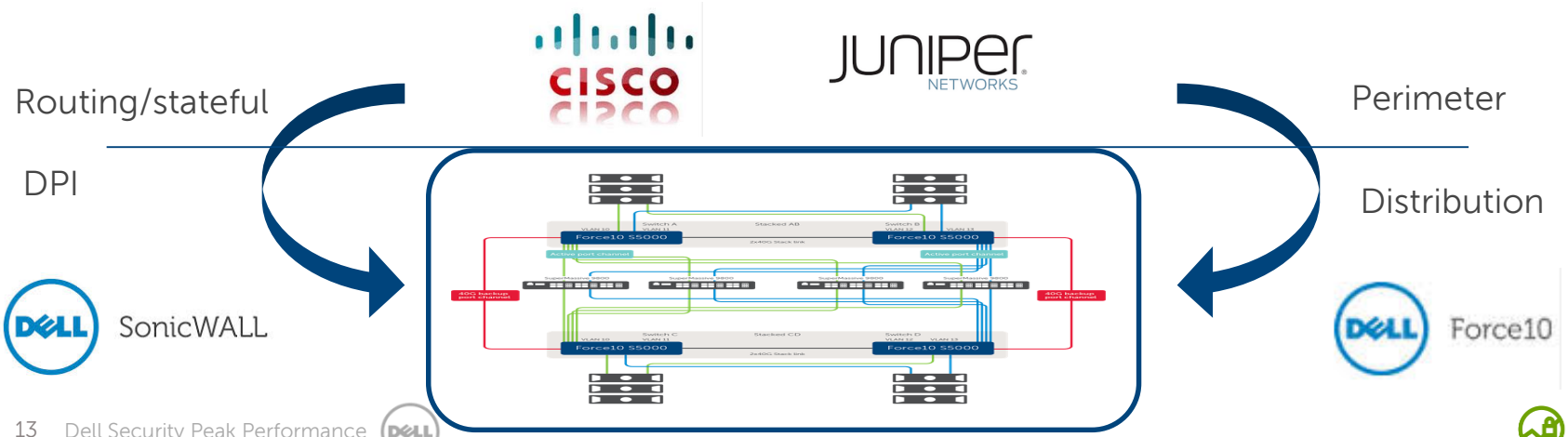


Figure 1 Traffic model evolution in data centers

- Security reasons
  - Scalability, commoditization of 10/40 GbE
  - Inspection at the Distribution/Core layer not just Access Layer (Perimiter)
  - East to West inspection
  - Malware Lifecycle (lateral movements, exfiltration, reconnaissance)

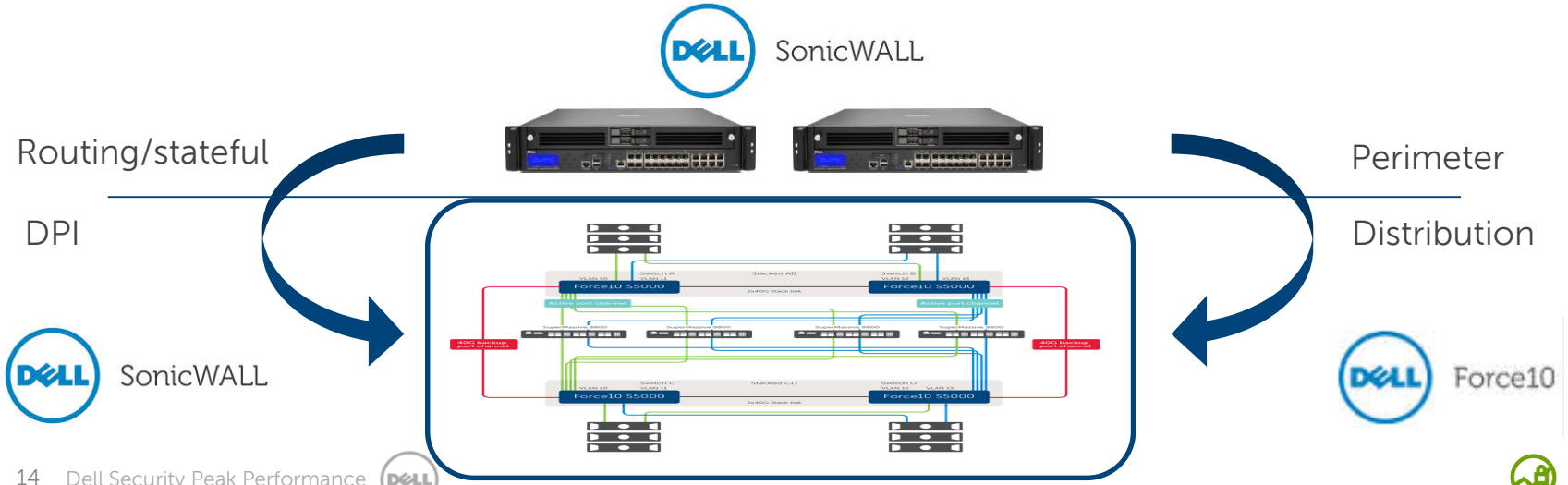
Firewall sandwich deployment options for all customer scenarios: **The classic firewall sandwich**

- Customer currently owns and opts to keep their existing firewall infrastructure
  - Usually a Cisco or Juniper solution deployed at the perimeter for stateful packet inspection, routing and NAT
  - Would like to add deep packet inspection (DPI)
  - Non-intrusive deployment, highly resilient design



Firewall sandwich deployment options for all customer scenarios: **The double firewall sandwich**

- Customer would like to completely replace their existing firewall infrastructure
- Requires routing and DPI solutions
  - While maintaining high performance and superb resiliency
  - Dell SonicWALL NGFW high-availability pair at the perimeter, providing Layer 3 services
  - Classic firewall sandwich providing all DPI services



Firewall sandwich deployment options for all customer scenarios: **The open firewall sandwich**

- Customer would like to replace their firewall solution with a Dell SonicWALL solution, but would prefer to leverage their existing switch infrastructure
  - Requires enormous amount of NAT and concurrent connections
  - Third-party Layer 3 network load balancer needed

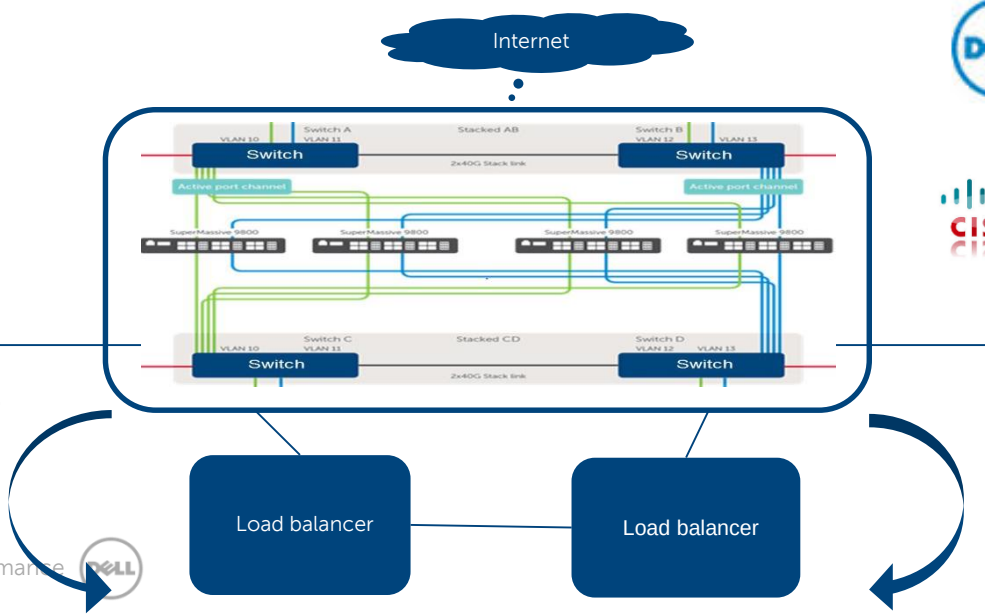
SonicWALL

Routing/stateful

DPI

Link aggregation

L3 load balancer



Force10

or

ARISTA

cisco

juniper NETWORKS

Perimeter

Distribution

Array NETWORKS

A10 Networks

f5 NETWORKS

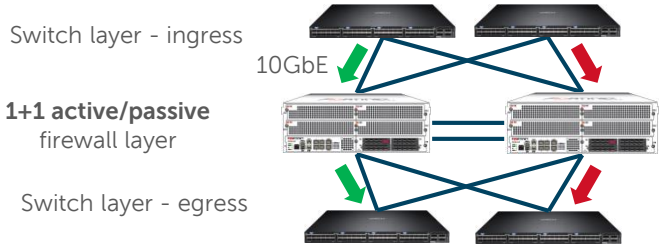
How far does it scale?

Cluster of firewall blades configuration (Wiremode, NATed/routed)							
Firewall mode	Security services	Platform	Dell F10 switches	Throughput	DPI-SSL throughput	TCP connection per seconds	Max. concurrent TCP connections
	IPS + App Intelligence	SM9400 (16 units)	S5000, S6000	160G (120G recommended for redundancy)	40G	2.2 million	24 million
Wiremode	IPS + App Intelligence	SM9800 (16 units)	S5000, S6000	320G (240G recommended for redundancy)	80G	4.5 million	40 million
	IPS, GAV, ASPY and Application Intelligence	SM9800 (16 units)	S5000, S6000	160G (120G recommended for redundancy)	80G	4.5 million	40 million
NATed/routed	IPS, GAV, ASPY and Application Intelligence	SM9400 (16 units) + 4 X SM9800	S5000, S6000	120G	40G	1.1 million	12 million
	IPS, GAV, ASPY and Application Intelligence	SM9800 (16 units) + 4 X SM10800	S5000, S6000	120G	80G	1.6 million	40 million

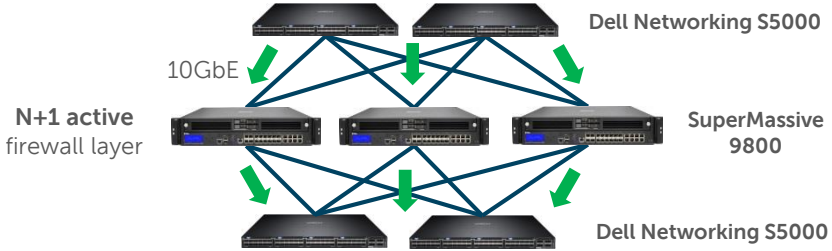
Transform NGFW economics and data center agility

Dell can deliver **cost savings** up to **85 percent***

Classic deployment



Dell firewall sandwich



* All pricing and performance figures taken from published information for comparable model or configurations.

References

Challenge

- Stateful firewall
- Research network has historically been open, leading to exposure and risk
- Growth, cloud initiatives and device proliferations
- Bandwidth increases to support 40Gbps and ultimately 100Gbps networks

Results

A network-based model for scaling a NGFW architecture to support 40Gbps to 100Gbps DPI performance

Resources

Dell Network Security: A Super Massively Scalable Network Firewall

Overview

As Network Security requirements have evolved, the response has been to scale up hardware to meet performance requirements. The trade-off has become increasingly larger, more complex, power-hungry, virtualized systems that are expensive to purchase and operate while increasing the impact in failure modes and providing a single attack point for denial-of-service and other firewall evasion techniques.

This document will describe a network-based model for scaling a Next Generation Firewall (NGFW) to approach or surpass existing or forthcoming solutions while providing increased performance, better TCO and increased resiliency.

How Massive Is Super?

Dell SonicWALL Network Security platforms employ a patented multi-core network processor architecture and Reassembly-free Deep Packet Inspection (RFDPI) engine. Together with our Cloud-assist Gateway Anti- (50%). Furthermore, denial-of-service or other attacks can be easily targeted to this single point, increasing the likelihood of failures. This is not a winning combination.

Using a network-based architecture, non-massive (1/2U) standard NGFW platforms can be deployed to scale infinitely, with similar or better TCO, better performance and increased resiliency to both failures and attacks. A fully-meshed L2 (transparent) architecture can consist of up to 16 NGFW devices all fully-active, providing up to **640Gbps of deep packet inspection** with typical failure modes that only impact n-1 of overall capacity.

There are additional benefits to the model, including the freedom to choose components based on price/performance, availability or other preferences. With this architecture, the devices don't' have to be massive to scale massively.

Upon Deeper Inspection

For the purposes of this paper, the Network-based firewall will be deployed in transparent (Layer 2) mode. This validated and supported architecture consists of Dell Networking S-series ingress/egress layer and Dell SonicWALL SuperMassive 9000 series security layer platforms. The table below identifies which security layer services are supported in Layer 2 vs. Layer 3 modes:

	Layer 2				Layer 3
	Bypass Mode	Inspect Mode	Secure Mode	Tap Mode	NAT, Route Modes
Active/Active Clustering *	No	No	No	No	Yes

Making a FireWALL Sandwich

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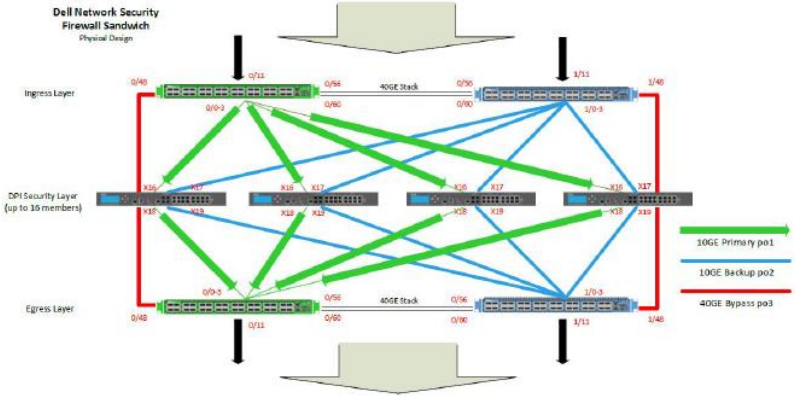
Firewall Sandwich Reference Architecture



Switching Layer Configuration Script

Assumption: Dell Networking S48x0, S5000 or S6000 switches in two stacks of two switches (Ingress and Egress layer stacks). 10GE interfaces used for ingress/egress and Firewall connectivity. 40GE interfaces used for stacking and bypass connectivity. Physical interfaces 0-11 (10GE) and 48, 56, 60 (40GE) used on each switch. Two 40GE interfaces remain available per stack for ingress/egress connectivity.

Thank you